Nalluri Sanjay

B.Tech (Hons.) Industrial and Systems Engineering, Minor in Computer Science, IIT Kharagpur
+91 74169 45107 | sanjaynalluri86@gmail.com | linkedin.com/in/sanjaynalluri2405 | github.com/SanjayNalluri

Education

B.Tech (Hons.), Industrial and Systems Engg.	IIT Kharagpur	2024 – 2028	CGPA 7.59/10
Class XII (IPE)	Narayana Junior College	2024	95.7%
Class X (SSC)	Montessori High School	2022	94.3%

Projects

Warehouse Demand Forecasting and Staffing Oct 2026
Personal Project | Python, LightGBM, SimPy, Rockwell Arena | GitHub | Live App
- Forecasted daily picking workload 7 days ahead from 1 million order lines of a UK wholesaler (WAPE 22.1%)
- Converted forecasts into weekly staffing plans; staffing to the P75 forecast cut late work from 6.4% to 0.9%
- Simulated shift patterns in SimPy and validated the model in Rockwell Arena
- Showed that a split shift reduces late order lines from 9.4% to 1.4% at 6.5% lower labour cost

DemandIQ: Demand Forecasting and Inventory Planning Oct 2026
Personal Project | Python, statsmodels, XGBoost | GitHub
- Forecasted daily demand for 20 Walmart grocery items (M5 dataset) using Holt-Winters and XGBoost
- Detected and corrected 3,797 stockout days so that lost sales were not treated as low demand
- Reduced forecast error (WAPE) from 30.9% to 23.3% compared with a seasonal naive baseline
- Designed (Q, R) reorder policies that needed 22.5% less safety stock at a 95% service level

DeliveryLens: On-Time Delivery Analysis Oct 2026
Personal Project | SQL Server, Power BI | GitHub
- Cleaned 96,018 e-commerce orders in SQL Server and built a 3-page Power BI dashboard with 12 DAX measures
- Found that 98 sellers (3.4%) caused half of all late orders and that late deliveries rose to 15.1% in peak months

HubShift: Warehouse Network Design for India Oct 2026
Personal Project | Python, PuLP, Streamlit | GitHub | Live App
- Formulated a facility location MILP to select warehouses from 27 candidate sites serving 52 Indian cities
- Redesigned the network for the post-GST tax regime, reducing modelled logistics cost by 22–28%

LastMile Delhi: Delivery Route Planning Oct 2026
Personal Project | Python, Google OR-Tools | GitHub
- Solved a vehicle routing problem with time windows for 92 Delhi supermarkets under city no-entry rules
- Estimated the cost of the no-entry rules at 2 extra trucks and about 8% higher delivery cost

Ops Chat: Text-to-SQL Assistant Oct 2026
Personal Project | Python, Groq LLM, FastAPI, React | GitHub | Live App
- Built an LLM chatbot that answers delivery questions by writing SQL, with a check that blocks any data changes
- Achieved 86.7% accuracy on 60 unseen test questions; deployed on Vercel

Technical Skills

Supply Chain and OR:	Demand forecasting, inventory policies, facility location, vehicle routing, simulation
Languages:	Python, SQL, C
Libraries:	pandas, NumPy, statsmodels, XGBoost, LightGBM, PuLP, Google OR-Tools, SimPy
Tools:	MS Excel, Power BI, Tableau, Rockwell Arena, IBM CPLEX, Minitab, Git

Relevant Coursework

Supply Chain Management, Management of Inventory Systems, Production Planning and Control, Simulation, Quality Design and Control, Engineering Economy, Operations Research I and II, Probability and Statistics

Certifications

- Professional Certification Program in AI and Data Science, DRISHTI CPS IIT Indore (Intellipaat) Sep 2026
- Microsoft SQL Certification Training, Intellipaat Sep 2025
- Deloitte Data Analytics Job Simulation, Forage May 2025

Achievements and Activities

- Secured All India Rank 1409 in JEE Advanced 2024
- Led a team of 6 to build a VR banking simulator in Unity for Meta Quest (Work System Design Lab)
- Won the Gold Medal at the Illumination event as part of the MMM Hall team, IIT Kharagpur